



Class 06 - Maths (NEW)

Maximum Marks: 40

- All questions are compulsory.
- Marks are provided alongside each question.
- Chapters included are:
 - Patterns in Mathematics
 - Lines and Angles
 - Number Play

- | | | |
|-----|--|-----|
| (a) | Which of the following is a correct representation of an obtuse angle? | [1] |
| | a) 135° b) 90° | |
| | c) 45° d) 180° | |
| (b) | What is the digit sum of the number 731? | [1] |
| | a) 9 b) 12 | |
| | c) 10 d) 11 | |
| (c) | The sequence 1, 3, 6, 10, 15, ... is called as | [1] |
| | a) Counting Numbers b) Squares | |
| | c) Odd Numbers d) Triangular Numbers | |
| (d) | Which angle is larger? | [1] |
| | a) 80° b) 45° | |
| | c) 60° d) 120° | |
| (e) | Assertion (A): The number 1 is neither prime nor composite. | [1] |
| | Reason (R): The number 1 has only one factor. | |
| | a) Both A and R are true and R is the correct explanation of A. | |
| | b) Both A and R are true but R is not the correct explanation of A. | |
| | c) A is true but R is false. | |
| | d) A is false but R is true. | |

- (a) A _____ has no length, breadth, or thickness and is usually represented by a dot. **[1]**
- (b) The Kaprekar process using the number 2111 reaches 6174 in _____ steps. **[1]**
- (c) In a complete graph, the number of lines (edges) follows the pattern of _____. **[1]**

numbers.

- (d) A _____ is a part of a line that has two endpoints.

[1]

3. **State whether each of the following statements is True or False:**

[3]

- (a) Geometry is the branch of mathematics that studies patterns in shapes. [1]
(b) An angle is formed by two rays that do not share a common starting point. [1]
(c) Supercells are grids that help reveal patterns in numbers when filled with sequences like multiples. [1]

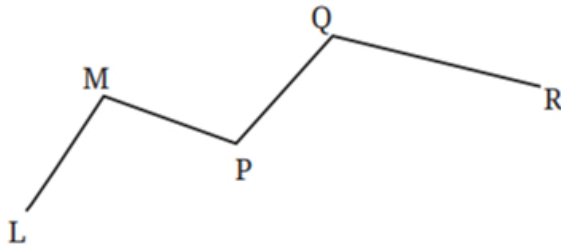
Section B

4. **Answer any THREE of the following questions:**

[3]

- (a) Name the line segments in Fig. Which of the five marked points are on exactly one of the line segments? Which are on two of the line segments?

[1]



- (b) Construct the Collatz sequence starting with 22. [1]
(c) Why are 1, 4, 9, 16, 25, 36, ... called square numbers? [1]
(d) A ship sailing in river Jhelam moves towards east. If it changes to north, through what angle does it turn? [1]

5. **Answer any TWO of the following questions:**

[4]

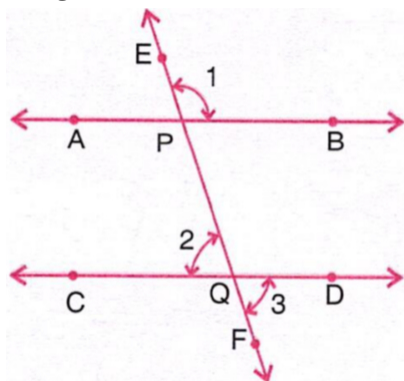
- (a) Which number sequence is formed by counting the number of lines in each shape in the sequence of complete graphs? [2]
(b) What is the use of a compass in geometry? [2]
(c) What is the sum of the smallest and the largest 6-digit palindromes? [2]

6. **Answer any TWO of the following questions:**

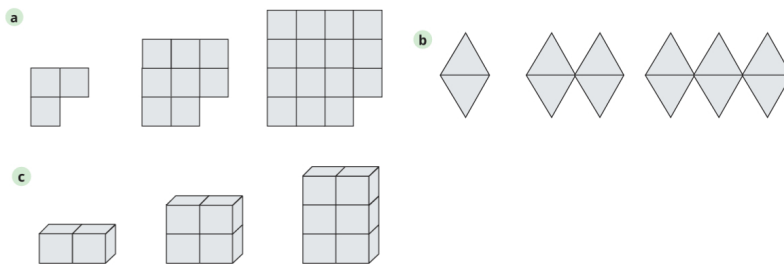
[6]

- (a) In figure, write another name for:

[3]



- i. $\angle 1$
ii. $\angle 2$
iii. $\angle 3$
(b) Find out the digit sums of all the numbers from 40 to 70. [3]
(c) What will be 6th shape in the patterns shown below? [3]



7. **Answer any TWO of the following questions:** [8]
- (a) How many supercells are possible for different number of cells in a one dimensional slid? Do you notice any pattern? [4]
- (b) Observe the pattern in the following and fill in the blanks: [4]
- $$9 \times 9 + 7 = 88$$
- $$98 \times 9 + 6 = 888$$
- $$987 \times 9 + 5 = 8888$$
- $$9876 \times 9 + 4 = \dots$$
- $$98765 \times 9 + 3 = \dots$$
- $$987654 \times 9 + 2 = \dots$$
- $$9876543 \times 9 + 1 = \dots$$
- (c) Draw rough diagrams of two angles such that they have [4]
- One point in common.
 - Two points in common.
 - Three points in common.
 - Four points in common.
 - One ray in common.

Section C

8. **Read the following text carefully and answer the questions that follow:** [4]
- In a school, students are arranging chairs in the auditorium for an event. They noticed that arranging chairs in a triangular pattern helps save space. The number of chairs in each row forms a sequence: 1, 3, 6, 10, 15, ... These numbers are called triangular numbers.
- What is the next triangular number after 15? (1)
 - How many chairs are there in the 4th row of the triangular arrangement? (1)
 - Explain how triangular numbers are formed. (2)
- OR**
- Draw a diagram to show the arrangement of chairs for the first 4 triangular numbers. (2)
9. **Read the following text carefully and answer the questions that follow:** [4]
- The Ashoka Chakra on the Indian flag has 24 spokes. Each spoke is equally spaced, forming equal angles between them. Ravi wants to calculate the angle between two adjacent spokes. Since a circle has 360 degrees, dividing it by 24 spokes gives 15 degrees between each spoke. Ravi calculates that the angle between two adjacent spokes is 15 degrees. This helps him understand the symmetry of the Ashoka Chakra.
- How many degrees are in a full circle? (1)
 - What is the angle between two adjacent spokes of the Ashoka Chakra? (1)

c. Calculate the angle between two spokes that are 5 spokes apart. **(2)**

OR

If the Ashoka Chakra had 12 spokes instead of 24, what would be the angle between two adjacent spokes? **(2)**